

Problem 1

Derive from the inverse twist problem equations (p. 34 of notes) the simplified forward and inverse equations for the twist parameterization in the special case of parallel displacement $\mathbf{d}$ with the rotation axis $\hat{\mathbf{k}}$.

$$\mathbf{d} = ((\mathbf{I} - \mathbf{R}) \left[\hat{\mathbf{k}} \right]_{\times} + \theta \hat{\mathbf{k}} \hat{\mathbf{k}}^{\top}) \mathbf{v}, \quad (1)$$

$$\mathbf{v} = \left(\frac{\sin \theta}{2(1 - \cos \theta)} \mathbf{I} + \frac{2(1 - \cos \theta) - \theta \sin \theta}{2\theta(1 - \cos \theta)} \hat{\mathbf{k}} \hat{\mathbf{k}}^{\top} - \frac{1}{2} \left[\hat{\mathbf{k}} \right]_{\times} \right) \mathbf{d}, \quad (2)$$

$$\mathbf{R} \hat{\mathbf{k}} = \hat{\mathbf{k}}, \quad \mathbf{R}^{\top} \hat{\mathbf{k}} = \hat{\mathbf{k}} \Rightarrow \hat{\mathbf{k}}^{\top} (\mathbf{I} - \mathbf{R}) = \mathbf{0}, \quad (3)$$

$$\text{Since } \hat{\mathbf{k}}^{\top} (\mathbf{I} - \mathbf{R}) = \mathbf{0} \text{ by (3), } (\mathbf{I} - \mathbf{R}) \mathbf{w} \perp \hat{\mathbf{k}} \quad \forall \mathbf{w} \Rightarrow (\mathbf{I} - \mathbf{R}) \left[\hat{\mathbf{k}} \right]_{\times} \mathbf{v} \perp \hat{\mathbf{k}}, \quad (4)$$

$$\mathbf{d} \parallel \hat{\mathbf{k}} \Rightarrow (\mathbf{I} - \mathbf{R}) \left[\hat{\mathbf{k}} \right]_{\times} \mathbf{v} = \mathbf{0}, \quad (5)$$

With (5), $\theta \neq 0 \pmod{2\pi}$: $(\mathbf{I} - \mathbf{R})$ invertible on $\hat{\mathbf{k}}^{\perp}$, and $\left[\hat{\mathbf{k}} \right]_{\times} \mathbf{v} \in \hat{\mathbf{k}}^{\perp}$

$$\Rightarrow \hat{\mathbf{k}} \times \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v} \parallel \hat{\mathbf{k}}, \quad (6)$$

$$\mathbf{v} = v \hat{\mathbf{k}}, \quad \mathbf{d} = d \hat{\mathbf{k}}, \quad (7)$$

$$d \hat{\mathbf{k}} = \theta \hat{\mathbf{k}} \hat{\mathbf{k}}^{\top} (v \hat{\mathbf{k}}) = \theta v \hat{\mathbf{k}}, \quad (8)$$

$$\boxed{d = \theta v}, \quad (9)$$

$$v \hat{\mathbf{k}} = \left[\frac{\sin \theta}{2(1 - \cos \theta)} + \frac{2(1 - \cos \theta) - \theta \sin \theta}{2\theta(1 - \cos \theta)} \right] d \hat{\mathbf{k}}, \quad (10)$$

$$\frac{\theta \sin \theta + 2(1 - \cos \theta) - \theta \sin \theta}{2\theta(1 - \cos \theta)} = \frac{2(1 - \cos \theta)}{2\theta(1 - \cos \theta)} = \frac{1}{\theta}, \quad (11)$$

$$\boxed{v = \frac{d}{\theta}}. \quad (12)$$

Problem 2

Sketch the zero-angle configuration from the following DH table.

Link	a	d	α	θ
1	1	1	$\pi/2$	*
2	0.5	1	$-\pi/2$	*
3	0	0	0	*
4	1	0	$\pi/2$	*
5	0	0	$-\pi/2$	*
6	0	0	$\pi/2$	*

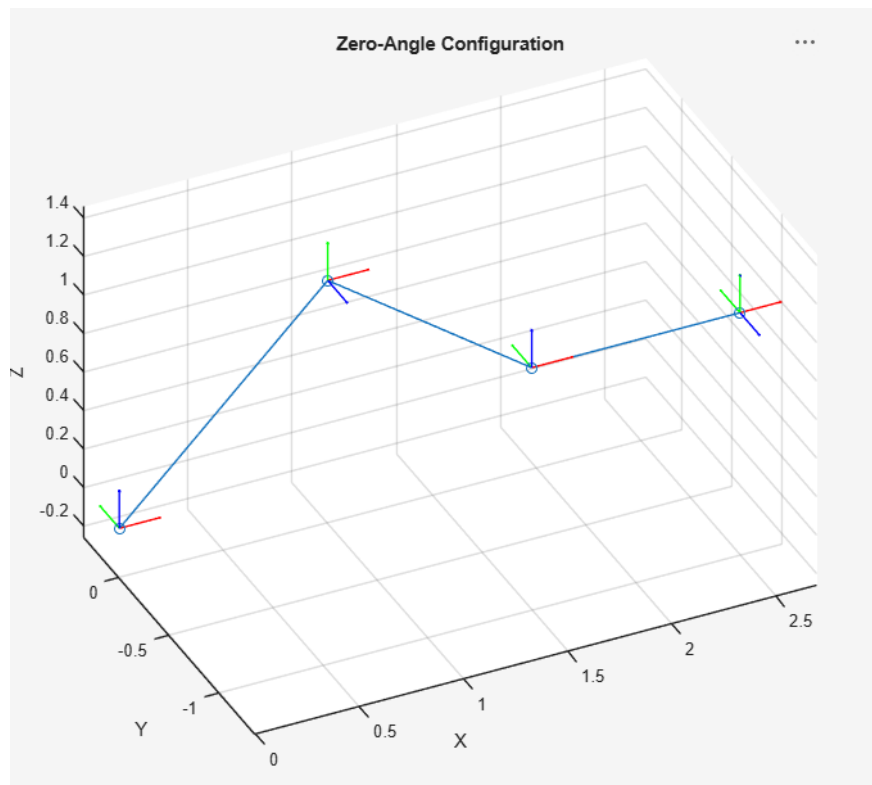


Figure 1: Zero-angle configuration for the manipulator of Problem 2. Frames 2 and 3 share an origin, and frames 4, 5, and 6 share one, so their axes overlap in the plot.

Problem 3

Sketch the zero-angle configuration from the following DH table.

Link	a	d	α	θ
1	1	$\star$	$\pi/2$	0
2	1	$\star$	$\pi/2$	$\pi/2$
3	1	$\star$	0	0

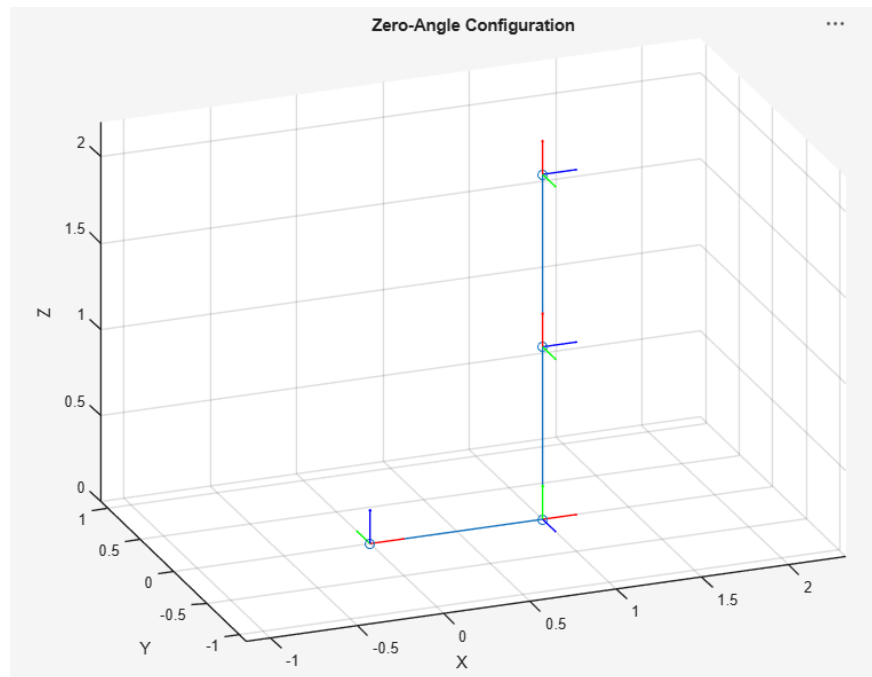


Figure 2: Zero-angle configuration for the manipulator of Problem 3.

Problem 4

For the manipulator of Problem 2:

- (a) What is the homogeneous transformation 0T_6 if all joint angles are 0?
- (b) What is the homogeneous transformation 0T_6 if all joint angles are $\pi/6$?

Part (a)

Evaluating the DH table of Problem 2 with all $\theta_i = 0$ (see [Problem4.m](#), [Appendix A](#)):

$${}^0T_6 = \begin{bmatrix} 1 & 0 & 0 & 2.5 \\ 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Part (b)

Evaluating the same DH table with all $\theta_i = \pi/6$ (see [Problem4.m](#), [Appendix A](#)):

$${}^0T_6 = \begin{bmatrix} -0.6808 & -0.3460 & 0.6456 & 1.6830 \\ 0.6456 & -0.6998 & 0.3058 & 0.8170 \\ 0.3460 & 0.6250 & 0.6998 & 1.5000 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Problem 5

For the manipulator of Problem 3:

- (a) What is 0T_3 when $d_1 = 1$ and all other joints are 0?
- (b) What is 0T_3 when $d_2 = 1$ and all other joints are 0?
- (c) What is 0T_3 when $d_3 = 1$ and all other joints are 0?

Part (a)

Evaluating the DH table of Problem 3 with $d_1 = 1$, $d_2 = d_3 = 0$ (see [Problem5.m](#), [Appendix A](#)):

$${}^0T_3 = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Part (b)

Evaluating the same DH table with $d_2 = 1$, $d_1 = d_3 = 0$ (see [Problem5.m](#), [Appendix A](#)):

$${}^0T_3 = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Part (c)

Evaluating the same DH table with $d_3 = 1$, $d_1 = d_2 = 0$ (see [Problem5.m](#), [Appendix A](#)):

$${}^0T_3 = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Problem 6

Derive the DH parameters for the RRPR manipulator shown in the figure below.

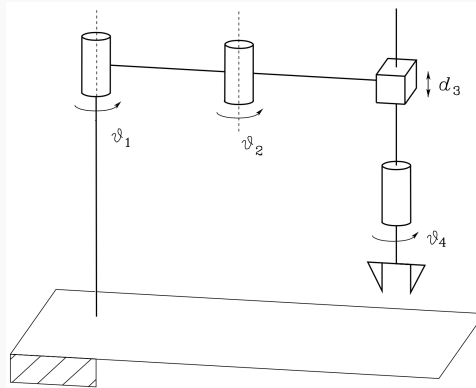


Figure 3: RRPR manipulator.

Frame assignment:

- Frame 0: z_0 up along the joint 1 axis; origin at arm height.
- Frame 1: z_1 up along the joint 2 axis (parallel to z_0); x_1 along link 1, from joint 1 to joint 2.
- Frame 2: z_2 down along the joint 3 (prismatic) axis; x_2 along link 2, from joint 2 to joint 3.
- Frame 3: z_3 down along the joint 4 axis (parallel to z_2); x_3 along x_2 , since $a_3 = 0$.
- Frame 4: end-effector frame at the tip, z_4 down along z_3 .

Parameter definitions:

- a_1 : link 1 length, horizontal offset from the joint 1 axis to the joint 2 axis.
- a_2 : link 2 length, horizontal offset from the joint 2 axis to the joint 3 axis.
- d_1 : vertical offset from frame 0 to frame 1. We take frame 0 at arm height (not the base plate), so $d_1 = 0$, matching the standard textbook version of this manipulator.
- d_4 : fixed vertical offset from frame 3 to the end-effector tip.
- $\alpha_2 = \pi$: z_2 (prismatic axis, pointing down) is antiparallel to z_1 (pointing up), a π twist about x_2 .

Link	a	d	α	θ
1	a_1	0	0	*
2	a_2	0	π	*
3	0	*	0	0
4	0	d_4	0	*

Appendix A: Code

Problem 2

Listing 1: Problem2.m

```

1 % Problem2: Sketch the zero-angle configuration from the given DH table.
2 %
3 % Arael Anaya
4 % 10920967
5 % MEGN544
6 % 09/28/2026
7
8 clear; close all; clc;
9
10 addpath(genpath('.../.../Functions'));
11
12 %% DH table (Problem 2)
13 % Link    a      d      alpha      theta
14 DH = [      1      1      pi/2      0 ; % Link 1
15        0.5      1     -pi/2      0 ; % Link 2
16        0      0      0      0 ; % Link 3
17        1      0      pi/2      0 ; % Link 4
18        0      0     -pi/2      0 ; % Link 5
19        0      0      pi/2      0 ]; % Link 6
20
21 n = size(DH, 1);
22
23 T_list = zeros(4, 4, n);
24 for i = 1:n
25     T_list(:,:,i) = dhTransform(DH(i,1), DH(i,2), DH(i,3), DH(i,4));
26 end
27
28 T01 = T_list(:,:,1);
29 T02 = T01 * T_list(:,:,2);
30 T03 = T02 * T_list(:,:,3);
31 T04 = T03 * T_list(:,:,4);
32 T05 = T04 * T_list(:,:,5);
33 T06 = T05 * T_list(:,:,6);
34
35 figure('Name', 'Problem 2: Zero-Angle Configuration');
36 hold on; grid on; axis equal;
37 xlabel('X'); ylabel('Y'); zlabel('Z');
38 title('Zero-Angle Configuration');
39 view(3);
40
41 origins = [0 0 0; T01(1:3,4)'; T02(1:3,4)'; T03(1:3,4)'; T04(1:3,4)'; T05
42            (1:3,4)'; T06(1:3,4)'];
43
44 plot3(origins(:,1), origins(:,2), origins(:,3), '-o');
45
46 axisLen = 0.2;
47 frames = {eye(4), T01, T02, T03, T04, T05, T06};
48 for i = 1:numel(frames)
49     T = frames{i};

```

```
48     p = T(1:3,4);
49     xAxis = T(1:3,1);
50     yAxis = T(1:3,2);
51     zAxis = T(1:3,3);
52     quiver3(p(1), p(2), p(3), xAxis(1), xAxis(2), xAxis(3), axisLen, 'r');
53     quiver3(p(1), p(2), p(3), yAxis(1), yAxis(2), yAxis(3), axisLen, 'g');
54     quiver3(p(1), p(2), p(3), zAxis(1), zAxis(2), zAxis(3), axisLen, 'b');
55 end
```

Problem 3

Listing 2: Problem3.m

```

1 % Problem3: Sketch the zero-angle configuration from the given DH table.
2 %
3 % Arael Anaya
4 % 10920967
5 % MEGN544
6 % 09/28/2026
7
8 clear; close all; clc;
9
10 addpath(genpath('.../.../Functions'));
11
12 %% DH table (Problem 3)
13 % Link    a    d    alpha    theta
14 DH = [    1    0    pi/2    0    ;    % Link 1
15        1    0    pi/2    pi/2    ;    % Link 2
16        1    0    0    0    ;    % Link 3
17
18 n = size(DH, 1);
19
20 T_list = zeros(4, 4, n);
21 for i = 1:n
22     T_list(:,:,i) = dhTransform(DH(i,1), DH(i,2), DH(i,3), DH(i,4));
23 end
24
25 T01 = T_list(:,:,1);
26 T02 = T01 * T_list(:,:,2);
27 T03 = T02 * T_list(:,:,3);
28
29
30 figure('Name', 'Problem 3: Zero-Angle Configuration');
31 hold on; grid on; axis equal;
32 xlabel('X'); ylabel('Y'); zlabel('Z');
33 title('Zero-Angle Configuration');
34 view(3);
35
36 origins = [0 0 0; T01(1:3,4)'; T02(1:3,4)'; T03(1:3,4)'];
37 plot3(origins(:,1), origins(:,2), origins(:,3), '-o');
38
39 axisLen = 0.2;
40 frames = {eye(4), T01, T02, T03};
41 for i = 1:numel(frames)
42     T = frames{i};
43     p = T(1:3,4);
44     xAxis = T(1:3,1);
45     yAxis = T(1:3,2);
46     zAxis = T(1:3,3);
47     quiver3(p(1), p(2), p(3), xAxis(1), xAxis(2), xAxis(3), axisLen, 'r');
48     quiver3(p(1), p(2), p(3), yAxis(1), yAxis(2), yAxis(3), axisLen, 'g');
49     quiver3(p(1), p(2), p(3), zAxis(1), zAxis(2), zAxis(3), axisLen, 'b');
50 end

```

Problem 4

Listing 3: Problem4.m

```

1 % Problem4: OT6 at theta = 0 and pi/6
2 %
3 % Arael Anaya
4 % 10920967
5 % MEGN544
6 % 09/28/2026
7
8 clear; close all; clc;
9
10 addpath(genpath('.../.../Functions'));
11
12 DH = [1      1      pi/2      0      ;
13        0.5    1     -pi/2      0      ;
14        0      0      0          0      ;
15        1      0      pi/2      0      ;
16        0      0     -pi/2      0      ;
17        0      0      pi/2      0      ];
18
19 T = fkChain(DH);
20
21 figure('Name', 'Problem 4: Zero-Angle Configuration');
22 hold on; grid on; axis equal;
23 xlabel('X'); ylabel('Y'); zlabel('Z');
24 title('Problem 4: Zero-Angle Configuration');
25 view(3);
26
27 origins = squeeze(T(1:3,4,:))';
28 plot3(origins(:,1), origins(:,2), origins(:,3), '-o');
29
30 axisLen = 0.2;
31 for i = 1:size(T,3)
32     Ti = T(:, :, i);
33     p = Ti(1:3,4);
34     quiver3(p(1), p(2), p(3), Ti(1,1), Ti(2,1), Ti(3,1), axisLen, 'r');
35     quiver3(p(1), p(2), p(3), Ti(1,2), Ti(2,2), Ti(3,2), axisLen, 'g');
36     quiver3(p(1), p(2), p(3), Ti(1,3), Ti(2,3), Ti(3,3), axisLen, 'b');
37 end
38
39 disp('A:'); disp(T(:, :, end));
40
41 t = pi/6;
42 DH(:,4) = t;
43 T = fkChain(DH);
44 disp('B:'); disp(T(:, :, end));

```

Problem 5

Listing 4: Problem5.m

```

1 % Problem5: Effect of link offset d on OT3
2 %
3 % Arael Anaya
4 % 10920967
5 % MEGN544
6 % 09/28/2026
7
8 clear; close all; clc;
9
10 addpath(genpath('.../.../Functions'));
11
12 DH = [1      0      pi/2      0      ;
13        1      0      pi/2      pi/2;
14        1      0      0        0      ];
15
16 T = fkChain(DH);
17
18 figure('Name', 'Problem 5: Zero-Angle Configuration');
19 hold on; grid on; axis equal;
20 xlabel('X'); ylabel('Y'); zlabel('Z');
21 title('Problem 5: Zero-Angle Configuration');
22 view(3);
23
24 origins = squeeze(T(1:3,4,:))';
25 plot3(origins(:,1), origins(:,2), origins(:,3), '-o');
26
27 axisLen = 0.2;
28 for i = 1:size(T,3)
29     Ti = T(:, :, i);
30     p = Ti(1:3,4);
31     quiver3(p(1), p(2), p(3), Ti(1,1), Ti(2,1), Ti(3,1), axisLen, 'r');
32     quiver3(p(1), p(2), p(3), Ti(1,2), Ti(2,2), Ti(3,2), axisLen, 'g');
33     quiver3(p(1), p(2), p(3), Ti(1,3), Ti(2,3), Ti(3,3), axisLen, 'b');
34 end
35
36 d = 1;
37
38 DH = [1      d      pi/2      0      ;
39        1      0      pi/2      pi/2;
40        1      0      0        0      ];
41 T = fkChain(DH);
42 disp('A:'); disp(T(:, :, end));
43
44 DH = [1      0      pi/2      0      ;
45        1      d      pi/2      pi/2;
46        1      0      0        0      ];
47 T = fkChain(DH);
48 disp('B:'); disp(T(:, :, end));
49
50 DH = [1      0      pi/2      0      ;

```

```
51         1      0      pi/2      pi/2;  
52         1      d      0      0  ];  
53 T = fkChain(DH);  
54 disp('C:'); disp(T(:, :, end));
```

dhTransform.m

dhTransform.m calls the custom `rotZ` and `rotX` functions below (both take radians), not MATLAB's built-in `rotz/rotx` (which take degrees).

Listing 5: dhTransform.m

```
1 % dhTransform: Homogeneous transform from DH parameters.
2 %
3 % H = dhTransform(a, d, alpha, theta)
4 % Homogeneous transform from DH parameters.
5 %
6 % H = [4x4] homogeneous transform
7 %
8 % a = link length
9 % d = link offset
10 % alpha = link twist (rad)
11 % theta = joint angle (rad)
12 %
13 % Arael Anaya
14 % 10920967
15 % MEGN544
16 % 09/28/2026
17 function H = dhTransform(a, d, alpha, theta)
18     R = rotZ(theta) * rotX(alpha);
19     p = [a*cos(theta); a*sin(theta); d];
20
21     H = [R, p;
22          0, 0, 0, 1];
23 end
```

rotZ.m

Listing 6: rotZ.m

```
1 % rotZ: Rotation matrix about the Z axis.
2 %
3 % R = rotZ(theta)
4 % Rotation matrix about the Z axis.
5 %
6 % R = [3x3] rotation matrix
7 %
8 % theta = rotation angle (rad)
9 %
10 % Arael Anaya
11 % 10920967
12 % MEGN544
13 % 09/19/2026
14 function R = rotZ(theta)
15     c = cos(theta);
16     s = sin(theta);
17     R = [c -s 0;
18         s  c 0;
19         0  0 1];
20 end
```

rotX.m

Listing 7: rotX.m

```
1 % rotX: Rotation matrix about the X axis.
2 %
3 % R = rotX(theta)
4 % Rotation matrix about the X axis.
5 %
6 % R = [3x3] rotation matrix
7 %
8 % theta = rotation angle (rad)
9 %
10 % Arael Anaya
11 % 10920967
12 % MEGN544
13 % 09/19/2026
14 function R = rotX(theta)
15     c = cos(theta);
16     s = sin(theta);
17     R = [1 0 0;
18         0 c -s;
19         0 s  c];
20 end
```

fkChain.m

Listing 8: fkChain.m

```

1 % fkChain: Cumulative forward-kinematics transforms from a DH table.
2 %
3 % T = fkChain(DH)
4 % Cumulative forward-kinematics transforms from a DH table.
5 %
6 % T = [4x4x(n+1)] stack of transforms, T(:,:,1) = base frame,
7 %     T(:,:,i+1) = OTi for link i
8 %
9 % DH = [n x 4] table of [a, d, alpha, theta] rows
10 %
11 % Arael Anaya
12 % 10920967
13 % MEGN544
14 % 09/28/2026
15 function T = fkChain(DH)
16     n = size(DH, 1);
17     T = zeros(4, 4, n+1);
18     T(:,:,1) = eye(4);
19     for i = 1:n
20         Ti = dhTransform(DH(i,1), DH(i,2), DH(i,3), DH(i,4));
21         T(:,:,i+1) = T(:,:,i) * Ti;
22     end
23 end

```